

VAJRA

Visual Audio Joint-modal Rendering Architecture

VAJRA 2.0

Document 1 of 5

System Requirements and Sizing

What the host machine must provide, and why

WHO THIS IS FOR

Whoever specifies or approves the machine

WHAT IT COVERS

- Minimum and recommended GPU, VRAM, RAM and disk
- Per-module cost, so a smaller host can run a subset
- Why VRAM is the largest module, not the sum of them
- The automated preflight check that verifies all of it

THE FULL SET

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|--|---------------------------|
| ■ 1. System Requirements and Sizing | specifying the machine |
| 2. Deployment and Operations Guide | installing and running it |
| 3. Operator's User Manual | using it |
| 4. Technical Writeup | understanding it |
| 5. Verification and Test Documentation | maintaining it |

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Offline synthetic-media platform · for authorised research and capability development

1. The Short Answer

Two configurations. Pick by whether the host must run the two video-generation engines, which are far larger than everything else combined.

	All seven tabs	Six tabs (no Text → Video)
GPU	NVIDIA, CUDA 12 capable	NVIDIA, CUDA 12 capable
VRAM — minimum	24 GB	16 GB
VRAM — recommended	48 GB	24 GB
System RAM — minimum	64 GB	32 GB
System RAM — recommended	128 GB	64 GB
Free disk — minimum	400 GB	150 GB
Free disk — recommended	600 GB, NVMe	250 GB, NVMe
Operating system	Linux (Ubuntu 22.04 LTS or similar)	Linux (Ubuntu 22.04 LTS or similar)
Interpreters	Python 3.12 and 3.10	Python 3.12 and 3.10
Network	Once, during install only	Once, during install only

Note. Cards that meet the recommended line comfortably: NVIDIA L40S (48 GB), RTX 6000 Ada (48 GB), A100 (40 or 80 GB), H100 (80 GB). At the 24 GB minimum: RTX 4090, RTX 3090, A5000. Below 16 GB only Media Studio, which is CPU-only, is dependable.

2. Why VRAM Is the Largest Module, Not the Sum

This is the single most useful thing to know when specifying the machine, and it is a deliberate design decision rather than a coincidence.

The platform holds **one model in GPU memory at a time**. A single-occupant cache evicts the resident model before loading the next, and a process-wide lock serialises every GPU job, so two large models are never resident together. If a second request arrives while one is running, it waits.

The consequence for sizing: **you need enough VRAM for the largest single module you intend to run, not for all of them added up**. Seven tabs do not need seven models' worth of memory.

Tip. The same decision is why the platform degrades predictably rather than mysteriously. Two large models in VRAM at once fails late and unpredictably; a queue fails visibly and never corrupts a run.

3. Per-Module Cost

Use this to size a host for a subset, and to know which tab to blame if memory runs short.

Tab	GPU memory in use	Basis	Environment
07 · Media Studio	none — CPU only	by design	principal runtime
04 · Face Swap	4 – 6 GB	guidance	principal runtime
02 · Voice Generation	6 – 10 GB	guidance	venv_voice, venv_viitor

Tab	GPU memory in use	Basis	Environment
03 · Image Generation	8 – 12 GB	guidance; model CPU offload and VAE tiling are enabled in code	principal runtime
01 · Video Edit	16 – 24 GB	stated in preflight	venv_latentsync, venv_voice
06 · Image Edit	20 – 24 GB	guidance; model CPU offload enabled	venv_qwen
05 · Text → Video	24 GB and upward	guidance; sequential CPU offload enabled, which trades speed for memory	venv_ltx2, venv_wan

Caution. Figures marked **guidance** are engineering estimates from the model sizes and the offload strategy in the code, not measurements taken on this hardware. Confirm them on the host during the first week and correct this table — section 8 says how.

4. Why System RAM Matters More Than Usual

The two video engines and the image editor use **CPU offload**: weights are held in system RAM and streamed onto the GPU layer by layer as they are needed. That is what allows a 27-billion-parameter model to run on a 24 GB card at all.

The cost is that the model must fit in **system** RAM instead. A host with a large GPU and only 32 GB of RAM will fail on the video tabs while appearing to have plenty of GPU memory free — an unintuitive failure worth anticipating.

If you plan to run	System RAM
Media Studio, Face Swap, Image Generation, Voice	32 GB
Video Edit as well	64 GB
Image Edit and Text → Video as well	128 GB recommended, 64 GB workable but tight

5. Disk — Where the Space Goes

Model weights dominate. The three group totals below are **measured from the repository's own file sizes**; the environment and working figures are estimates, marked as such.

Component	Size	Basis
Weights — core modules (image, voice, face, lip-sync, editing)	42.8 GB	measured
Weights — identity-preserving video (Wan2.2-I2V-A14B)	88.5 GB	measured
Weights — synchronised audio-video (LTX-2.3)	126.2 GB	measured
Weights, all seven tabs	257.5 GB	sum of the above; the installer warns that <i>all</i> exceeds 250 GB
Isolated environments (8, each with its own CUDA-enabled tensor stack)	≈ 45 – 55 GB	estimate — confirm on install
Third-party engine sources (cloned at pinned versions)	≈ 5 GB	estimate
Working space — uploads, outputs, intermediate renders	20 GB and upward	grows with use; not self-clearing
Total, all seven tabs	≈ 330 GB	specify 400 GB minimum, 600 GB comfortable
Total, six tabs (no Text → Video)	≈ 115 GB	specify 150 GB minimum, 250 GB comfortable

The installer's own thresholds agree with this: it **fails** below 60 GB free, and **warns** below 120 GB that the video modules alone exceed 100 GB each.

Put the weights somewhere sensible

Weights live wherever VAJRA_MODELS points; by default that is `models/` inside the project. On a shared machine, point it at a volume several people can read, so 257 GB is fetched once rather than once per user:

```
export VAJRA_MODELS=/data/vajra/models
bash install.sh voice lipsync
```

6. Software Prerequisites

Two Python versions are required, and this is the least obvious prerequisite on the list. The voice and lip-sync stacks depend on packages that publish no wheels for 3.12, so they are built from 3.10 while everything else runs on 3.12. Missing 3.10 is the most common reason an otherwise healthy host fails to install.

```
sudo apt install python3.12 python3.12-venv python3.12-dev \
    python3.10 python3.10-venv python3.10-dev \
    ffmpeg git git-lfs util-linux
```

Requirement	Why
NVIDIA driver, CUDA 12 capable	every GPU module
Python 3.12	principal runtime and the newer module stacks
Python 3.10	voice and lip-sync stacks — no 3.12 wheels exist
ffmpeg	all media handling, in every module
git, git-lfs	cloning the third-party engines at install
flock (util-linux)	without it, parallel environment builds are serialised — slower, still correct
sudo or root	only for the two apt steps; everything else runs unprivileged

7. Network

Install time: internet is needed once, for Python packages, model weights and the third-party clones. **Run time:** none. The platform makes no external call once installed, and the interface is served on localhost.

Caution. If the host will **never** have internet, the procedure changes substantially and must be planned before the transfer. See the Deployment Guide, *Air-gapped installation* — in particular the fact that four of the seven tabs fetch their weights on first use rather than during install.

8. Verify Before You Commit — and Measure After

Do not take this document's word for it. The repository carries a preflight check that inspects the actual machine and reports what it found, what is needed, and what to do about it:

```
python3 setup/preflight.py
```

It checks the GPU and driver, both interpreters, ffmpeg, git, git-lfs, flock, privileges, free disk and whether Hugging Face is reachable. It exits non-zero on a blocking problem, so it can gate an automated build. It writes nothing — safe to run on a machine you have not committed to.

Then measure the figures marked *guidance*

During the first week, run each tab once while watching GPU memory, and replace the estimates in section 3 with observations:

```
nvidia-smi --query-gpu=memory.used,memory.total \
--format=csv -l 2
```

Two minutes of work converts this sheet from a specification into a record, and it is the first thing a future purchaser will ask for.